

Packet Loss Modeling and Forward Erasure Correction for LEO Satellite Networks

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Abstract—Low earth orbit (LEO) satellite networks are pivotal for sixth-generation (6G) wireless systems, yet their high-speed mobility induces frequent packet loss, causing severe head-of-line blocking delays under traditional retransmission mechanisms. While streaming forward erasure correction (FEC) can mitigate retransmissions, existing packet loss models fail to capture the unique dynamics of LEO networks, causing difficulties in the design and analysis of FEC schemes. This paper addresses this problem through the following contributions. First, based on real-world Starlink measurements, we reveal the inadequacy of conventional loss models such as those based on Markov chains. Second, we propose a Markovian arrival process (MAP) to model LEO packet loss. Using an expectation-maximization (EM) algorithm to fit Starlink traces, we demonstrate its superior accuracy over existing models. Third, based on MAP modeling, we show that the decoding delay of a typical streaming FEC scheme with fixed repair insertion intervals can be analyzed by approximating it as the busy period of a MAP/D/1 queue. Using matrix-analytic methods, we provide a numerical recipe to compute this delay. Simulations validate the precision of the model in predicting delay, offering practical guidelines for FEC design in LEO networks.

Index Terms—LEO satellite networks, forward erasure correction, packet loss modeling, Markovian arrival process, decoding delay analysis.

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I. INTRODUCTION

LOW earth orbit (LEO) satellite networks play an important role in the future sixth-generation (6G) wireless system [2]. Starlink, arguably the largest LEO constellation, has entered large-scale deployment and operation, with subscribers exceeding 5 million. However, due to factors such as the high speed mobility of satellites [3], [4], end-to-end connections of LEO satellite networks have shown frequent packet loss in real-life measurements. Under this condition, the traditional retransmission-based recovery mechanism, which would incur a head-of-line (HoL) blocking delay proportional to the round-trip time (RTT) of the connection, can lead to significant goodput fluctuations, adversely affecting the user experience of many applications [5].

Forward erasure correction (FEC) can effectively prevent retransmissions. The basic idea is to insert redundant repair packets among the packet stream, so that when packet loss occurs, the repair packets can be used to recover the lost packets. Block codes and streaming codes are common forms of FEC. A block code encodes k source packets in a window into a set of $n > k$ encoded packets, so that any k or slightly larger than k encoded packets may reconstruct the source data. Typical block code examples are Reed-Solomon codes and Raptor codes [6]. Streaming FEC codes [7], [8], also known as convolutional encoding, have an elastic encoding window. Each repair packet is a random linear combination of possibly all of the packets that have not been acknowledged. Compared to block codes, streaming coding does not need to group source packets into blocks beforehand and hence is more flexible, and may also achieve a lower delivery delay thanks to the expanding encoding window at an increased encoding/decoding computational cost. This paper focuses on the streaming FEC.

The advantage of FEC over retransmission comes from trading bandwidth for reduced delay. To achieve the optimal trade-off, it is crucial that the proportion and the relative positions of repair packets among the packet stream fit the underlying loss random process of the connection [9], [10], [11]. Otherwise, repair packets may be wasted or not be sufficient to timely recover lost packets. To meet this goal, two closely coupled tasks, namely the modeling of loss and FEC analysis, respectively, need to be accomplished. For the modeling task, a packet loss model that is both accurate and analytically tractable needs to be obtained. This requirement

is two-fold. First, the model should accurately capture the complex loss pattern of LEO links, which has been shown to be bursty and highly dynamic due to rapid satellite movement and frequent handovers [3], [12], at reasonable costs of collecting training data and computation. Second, and perhaps more importantly, the model must possess mathematical tractability and otherwise the subsequent analysis task would become infeasible to accomplish. Supposing that a tractable model is available, the analysis task evaluates key performance metrics for certain FEC schemes, and ultimately provides means for code parameters design.

In the literature, the Bernoulli model where loss events are independent is perhaps the simplest statistical models for the modeling task. The two-state Markov chain Gilbert model [13] extends from this independent assumption and has been the basis for a broad class of Markov-based loss models. However, it may not accurately characterize the loss behavior in real networks [14], [15]. The Gilbert-Elliott (GE) and 4-state Markov (4SM) models [16] are extensions of the Gilbert model, and have been implemented in network simulators/emulators such as NetEm [17]. In [18], the loss run-length model (LRL) further extends the Markov chain model to include more possible loss states, to capture burst loss. In addition, the receive-loss run-length (RLRL) model extends the receiving state of the LRL model. However, it remains unclear whether the aforementioned models fit the connection characteristics of LEO satellite networks, which exhibit unique dynamics not seen in terrestrial networks for which the models were developed. For the analysis task, which specifically is to characterize the FEC decoding delay, several of the above models have been investigated for certain FEC schemes. For example, in [8] and [19], the FEC delay with fixed and random repair insertion interval, respectively, under Bernoulli losses are analyzed by modeling the delay as the busy period of G/D/1 and Geom/Geom/1 queues. In [20] and [21], the delay is analyzed and then optimized for the delayed-feedback case, where the transmission is modeled as a partially observable Markov decision process. In [22] and [23], a streaming code of specific rate $1/2$ is proposed for interactive multimedia, where the delay is analyzed from an information-theoretic perspective under the GE model. In [24], Markovian burst loss, which is a special case of 4SM, is considered where the bursts occur following a Poisson arrival process and the burst size is random. The FEC delay is then modeled as the busy period of an $M^X/G/1$ queue.

Recently, machine learning (ML) based methods have attracted much attention for time series modeling and prediction [25], and some applications have been reported for modeling network losses or delay. For example, ML has been employed to predict packet losses [26], predict RTT of LEO networks [27], and generate synthetic loss traces for network simulations [28]. The methods, however, are difficult to be applied to meet the goal of this paper as they only consider the standalone modeling task. Besides the possibly high costs of collecting a large amount of training data and efforts devoted to training/tuning the ML models, a critical issue with the ML-based methods is the lack of mathematical tractability, which is key in the aforementioned analysis task. In other

words, even if an accurate loss model can be obtained using ML, the ultimate goal of guiding the design of FEC is difficult to be met.

Based on the above discussions, this paper makes the following contributions on packet loss modeling and FEC design for LEO satellite networks.

- First, based on real-life measurements from a Starlink testbed, the performance of fitting the loss traces using existing packet loss models is evaluated. We reveal that existing models, including the mentioned statistical models and several typical ML models, may not correctly characterize the loss pattern of a LEO satellite network.
- Second, we propose to use the Markovian arrival process (MAP) [29], a versatile random process that was popular for modeling Internet traffic [30], to model the LEO loss traces by treating each loss event as an arrival. We devise an expectation-maximization (EM) algorithm to fit the parameters of the model from Starlink loss traces and show that the model outperforms the existing models. This is an innovative application of MAP, which extends its established use in traffic modeling to packet loss modeling.
- Third, considering a streaming FEC scheme with fixed repair packet insertion interval, we show that the decoding delay assuming a MAP loss model can be approximately analyzed based on the busy period of a MAP/D/1 queue. A complete numerical recipe is devised using the matrix analysis method (MAM) [31], [32] to compute the busy period. Simulations show that the numerical method correctly predicts the delay. Such mathematical tractability enables the derivation of optimal FEC intervals under delay constraints, as demonstrated by an application case.

The remainder of the paper is organized as follows. Section II presents the system model, including the end-to-end connection model and streaming FEC encoding/decoding. The existing statistical packet loss models are reviewed in Section III, and then the MAP loss model is proposed in Section IV. In Section V, the FEC decoding delay under the MAP loss is analyzed via MAP/D/1 queue's busy period, and the numerical recipe is presented. Section VI evaluates the proposed models. Finally, several future directions of the work are discussed in Section VII and the paper is concluded in Section VIII.

Notations: Throughout the paper, if not otherwise stated, bold capital letters and lowercase letters are used to denote matrices and vectors, respectively, with $(\cdot)^T$ denoting matrix transpose. If not otherwise stated, vectors are row vectors. $[\cdot]_i$ and $[\cdot]_{i,j}$ denote the i -th and the (i,j) -th element of a vector and matrix, respectively.

II. SYSTEM MODEL

A. Leo End-to-End Connection Model

We consider an end-to-end connection model of LEO satellite networks as shown in Fig. 1(a), where a client connects to a remote server via LEO satellites using a user terminal (UT). The end-to-end connection traverses a ground station

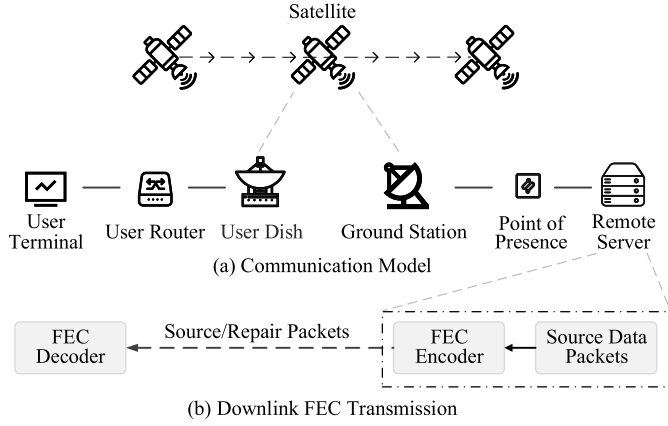


Fig. 1. End-to-end connection model of LEO satellite networks.

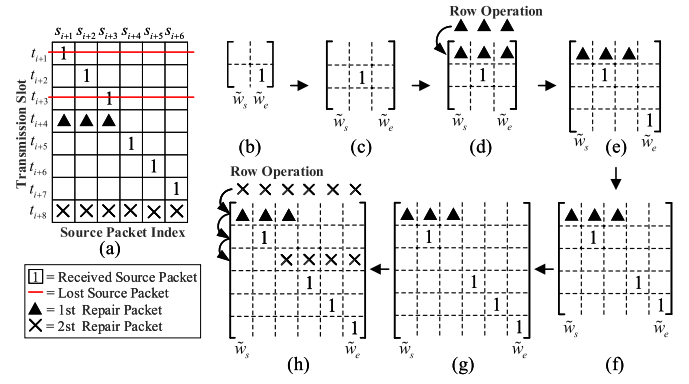
(GS) and a point of presence (PoP) to the Internet service provider, before finally reaching the remote server. Note that since each LEO satellite moves fast in space and only covers a limited area, the system needs to periodically switch the association of the UT to satellites. The transmission directions from the client (server) to the server (client) are referred to as *uplink* (*downlink*).

Over the end-to-end connection, we assume that equal-size packets are transmitted at a constant rate R_c . A packet may be lost at some point of the connection. For a sequence of packets over time $[0, T]$, we denote the packet loss trace $\mathbf{b} = \{b_k : k = 1, 2, \dots\}$, where $b_k = 1$ indicates that the k -th packet is lost and $b_k = 0$ otherwise. Given R_c , we can determine the time of the k -th loss and the interval between the k -th and $(k+1)$ -th loss among $\mathbf{b}$ as t_k and $y_k = t_k - t_{k-1}$, respectively. We denote $\mathbf{t} = \{t_k : 1 \leq k \leq n_t\}$ and $\mathbf{y} = \{y_k : 1 \leq k \leq n_t\}$, where n_t is the number of data points in the trace. Without loss of generality, we assume that $t_0 = 0$ and $t_{n_t} = T$.

B. Streaming FEC for In-Order Delivery

For ease of description, in this subsection we assume that the connection is a downlink transmission, as shown in Fig. 1(b), and the time is slotted such that sending each packet corresponds to one slot. The FEC scheme applies to uplink transmission as well. The transmission is to deliver packets from the server to the client in order. Let $\mathbf{s}_j, j = 0, 1, 2, \dots$ be the source packets to be sent. Each source packet is a row vector containing K elements from the finite field $\mathbb{F}_q$ of size q , i.e., each packet has $K \log_2 q$ bits. We consider the following streaming FEC scheme with constant repair insertion interval d , where a repair packet is sent after sending every $d - 1$ source packets. Denote the index (ID) of the most recently sent source packet as i_{seq} , initially $i_{\text{seq}} = -1$. When sending a source packet, $\mathbf{s}_{i_{\text{seq}}+1}$ is sent and $i_{\text{seq}} \leftarrow i_{\text{seq}} + 1$. When sending a repair packet, it is generated as a random linear combination of all the not-orderly delivered source packets so far. A repair packet of repair ID i can be expressed as

$$\mathbf{c}_i = \sum_{j=w_s}^{w_e} w_{i,j} \mathbf{s}_j,$$

Fig. 2. A decoding example with DW width 6 when $i_{\text{ord}} = i$.

where $w_{i,j}$ is the encoding coefficient randomly selected from $\mathbb{F}_q$, and w_s and w_e are the smallest and the largest ID of the source packets encoded in the repair packet, respectively. We refer to $[w_s, w_e]$ as the encoding window (EW), and $[w_{i,w_s}, \dots, w_{i,w_e}]$ as the encoding vector (EV) of $\mathbf{c}_i$. Such EV should be carried with the repair packet to enable decoding. In practice, to reduce overhead, usually only the repair ID is carried, which acts as the seed of an agreed pseudo-random number generator (PRNG) to recover the EV (e.g., [33]).

After leaving the server, the packets are transmitted through the lossy connection. At the client end, the lost source packets need to be recovered through decoding and delivered to the upper layer in their original order. Let i_{ord} represents the index of the most recent source packet delivered orderly. When the next received packet is an out-of-order packet, i.e., neither the source packet of ID $i_{\text{ord}} + 1$ nor a repair packet with $w_e = i_{\text{ord}} + 1$, the in-order delivery is interrupted, and the system enters an out-of-order period. The decoder is activated and attempts to recover the lost source packets to resume in-order delivery. Let $\tilde{w}_s \equiv i_{\text{ord}} + 1$, and $\tilde{w}_e$ be the maximum w_e 's among the EWs of the incoming repair packets until successful decoding. The range $[\tilde{w}_s, \tilde{w}_e]$ is referred to as the decoding window (DW).

In the decoding process, on-the-fly Gaussian elimination (OF-GE) [19] is employed to incrementally solve a system of linear equations $\mathbf{A}_c \times \mathbf{S} = \mathbf{C}$ based on the incoming packets. The decoding matrix $\mathbf{A}_c$ contains the encoding coefficients in its rows, while the information symbols are in the rows of $\mathbf{C}$. The incoming source packets after decoding activation are considered as special repair packets with only one nonzero encoding coefficient 1 corresponding to the packet itself. Performing forward row operations of the OF-GE algorithm on the EVs, an upper triangular $\mathbf{A}_c$ is maintained throughout the decoding, so that the decoder immediately knows when $\tilde{w}_e - \tilde{w}_s - 1$ innovative (linearly independent) EVs are received to complete decoding, i.e., when $\mathbf{A}_c$ becomes full rank.

Fig. 2 presents an example of decoding 6 packets during slots t_{i+1} to t_{i+8} , with $i_{\text{ord}} = i$ and $d = 4$. Fig. 2(a) provides an illustration of the packets. Specifically, source packets $\mathbf{s}_{i+1}, \dots, \mathbf{s}_{i+6}$ are transmitted in their original order at $t_{i+1}, t_{i+2}, t_{i+3}, t_{i+5}, t_{i+6}, t_{i+7}$, respectively. The strike-through lines indicate instances of packet loss. As shown in

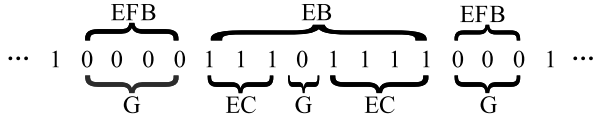


Fig. 3. Illustrative examples of loss trace modeling metrics; $\eta = 3$.

Fig. 2(b), as s_{i+2} is received, the decoder is activated since $i + 2 \neq i_{\text{ord}} + 1$. After receiving a repair packet, forward row operations are performed against the existing rows of $\mathbf{A}_c$ from top to bottom to populate the first row that has a zero diagonal element. Step (d) in Fig. 2 fills the EV (denoted by $\blacktriangle$) of the first repair packet in the first row of $\mathbf{A}_c$ immediately. When s_{i+4} , s_{i+5} , and s_{i+6} are received, the DW (i.e., the dimension of $\mathbf{A}_c$) naturally increases. In step (h), the second repair packet (whose EV is denoted by $\times$) is placed in the third row after processing against the first two rows of $\mathbf{A}_c$. At this moment, $\mathbf{A}_c$ is full-rank, and the decoding can be completed by performing backward row operations on $\mathbf{A}_c$. Packets within the DW are then delivered altogether to the upper layer, with new $i_{\text{ord}} = i + 6$ and in-order delivery resumed. In practice, the receiver may periodically report i_{ord} to the sender, such that the EW slides.

C. Performance Metrics

One goal of this paper is to fit the random process $\mathbf{b}$ (where 1 denotes packet loss and 0 denotes successful delivery), or equivalently $\mathbf{y}$, of LEO satellite networks using a mathematical model. To measure the accuracy of the modeling, we consider the following performance metrics:

- Gap distribution: Gap (G) is defined as a continuous 0's between two 1's in $\mathbf{b}$. The gap distribution $G(m_g)$ is defined as the probability distribution function for gaps of length m_g .
- Error cluster distribution: Error cluster (EC) is defined as consecutive 1's between two 0's in $\mathbf{b}$ [34]. The EC distribution $P(m_c)$ defines the probability distribution function for the occurrence of error clusters of length m_c or more.
- Error burst distribution: Error burst (EB) is defined as a sequence of error clusters with gaps greater than or equal to η length on both sides, or with gaps smaller than η between them. Where η is a predefined positive integer [35], [36]. The EB distribution $P_{EB}(m_e)$ is the probability distribution function of the error burst of length m_e .
- Error-free burst distribution: Error-free burst (EFB) is defined as a gap of length greater than or equal to η . The EFB distribution $P_{EFB}(m_e)$ is defined as the probability distribution function of an error-free burst of length m_e .

Note that while EFB derives from the Gap distribution, it directly quantifies the probability of sustained error-free communication exceeding η , offering a key reliability metric for LEO satellite communications. The above terms are illustrated in Fig. 3.

The purpose of streaming FEC is to reduce the in-order delivery delay over a lossy connection. When a source packet

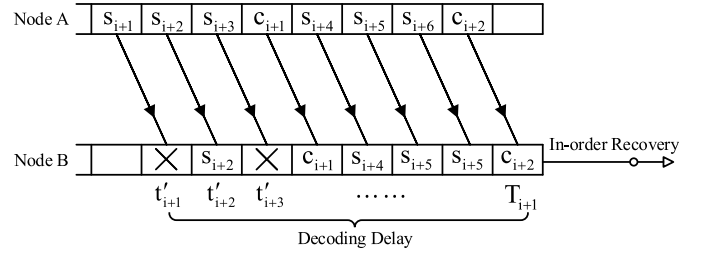


Fig. 4. Schematic illustration of decoding delay.

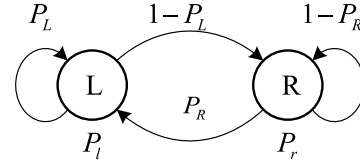


Fig. 5. Gilbert-Elliott model.

loss occurs, the orderly delivery halts. The out-of-order period lasts until the decoder receives at least an equal number of repair packets to that of the lost source packets during the activation. Such an out-of-order duration is equivalently the busy period of the decoder, and is also referred to as the decoding delay. The corresponding busy period of Fig. 2 is illustrated in Fig. 4. Let t'_{i+1} be the suppose-to-deliver time of the $i+1$ -th source packet if no loss had occurred, and T_{i+1} be the time the orderly delivery resumes. The decoding delay can be expressed as $T_{i+1} - t'_{i+1} + 1$ slots. The second goal of this paper is to characterize such decoding delay of the aforementioned streaming FEC scheme under the loss pattern of LEO satellite networks, by analyzing the decoder's expected busy period duration.

III. REVIEW OF EXISTING PACKET LOSS MODELS

In this section, we briefly review several existing models commonly used to characterize packet loss, and derive explicit formulas for fitting their parameters from the trace $\mathbf{b}$.

A. Existing Packet Loss Models

1) *Gilbert-Elliott Model*: Gilbert-Elliott model (GE) is a second-order hidden Markov chain model [13]. At its core, it assumes that the system state switches between two modes, namely the “good” state and the “bad” state (Fig. 5). The GE model assigns packet loss probabilities P_r , P_l to the receiving state (R) and the loss state (L), respectively.

To fit the model from $\mathbf{b}$, define the coefficients a_{ij} , $i, j \in \{0, 1\}$, with 0 and 1 corresponding to packet loss and successful delivery, respectively. These coefficients are calculated as

$$a_{i0} = \frac{P(ii)^2 - P(i)P(iii)}{P(11) - P(1)^2}, a_{i1} = \frac{P(iii) - P(i)P(ii)}{P(11) - P(1)^2},$$

where $i \in \{0, 1\}$. For $i = 1$, $P(1)$ is the empirical probability that 1 occurs in $\mathbf{b}$, $P(11)$ is the probability that the sequence ‘11’ appears in $\mathbf{b}$, and so on for $P(111)$; For $i = 0$, the definitions are similar. Based on a_{ij} 's, we can obtain

$$P_L = \frac{(a_{11} + a_{01})P_r - a_{11}}{(P_r - P_l)},$$

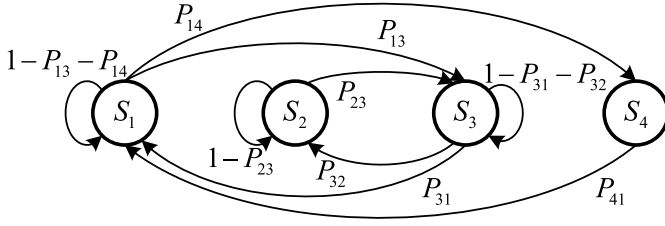


Fig. 6. 4-state Markov model.

$$P_R = 1 - \frac{a_{11} - (a_{11} + a_{01}) P_l}{(P_r - P_l)},$$

where P_L and P_R represent the transition probabilities to the loss state L from state L and state R , respectively. The state-dependent loss probabilities P_l and P_r are given by

$$P_l = \frac{1}{2} \left(\alpha + \sqrt{\alpha^2 - 4\beta} \right), P_r = \frac{1}{2} \left(\alpha - \sqrt{\alpha^2 - 4\beta} \right),$$

where

$$\alpha = \frac{a_{11} + a_{01} + a_{00} - a_{10} - 1}{a_{11} + a_{01} - 1}, \beta = -\frac{a_{10}}{a_{11} + a_{01} - 1}.$$

2) *4-State Markov Model*: Salsano et al. have proposed a 4-State Markov model (4SM) with two good and bad states [16] as shown in Fig. 6. The model has been implemented in Linux's *netem* emulator, and has been widely used to simulate packet losses for computer networks. The model treats a loss process as burst loss periods separated by gap periods. In each burst loss period, packet loss is very likely to happen with occasional successful reception. This is modeled by states S_2 and S_3 , which are the so-called bad states that correspond to occasional successful reception and loss during a burst loss period, respectively. In each gap period, successful reception is very likely, and only isolated (i.e., single) loss may occur during this period. This is modeled by S_1 and S_4 , which are the good states that correspond to a successful reception and an isolated loss, respectively.

Given the definition of states, it is clear that $P_{41} = 1$ so that only isolated loss would occur during a gap period. P_{13} and p_{31} are the transition probabilities between the gap and the burst loss periods. In [16], it is shown that the transition probabilities, along with P_{23} , P_{32} , P_{14} , can be determined by first modeling the loss trace as GE, which is to obtain P_L , P_R , P_l and P_r as described above. The overall packet loss probability and the isolated loss probability in 4SM can then be determined as¹

$$P_{\text{loss}} = \frac{P_r(1 - P_L) + P_l P_R}{1 + P_R - P_L} \text{ and } P_{\text{iso}} = P_r,$$

respectively. In addition, the mean burst length E_B and the density of losses ρ_{loss} during the burst period are

$$E_B = \frac{P_l + (1 - P_L)^2(1 - P_l)}{(1 - P_L)(P_l + (1 - P_L)(1 - P_l))},$$

$$\rho_{\text{loss}} = \frac{(P_l + (1 - P_L)(1 - P_l))^2}{P_l + (1 - P_L)^2(1 - P_l)}.$$

¹Note that the isolated loss in GE and that in 4SM are equivalent.

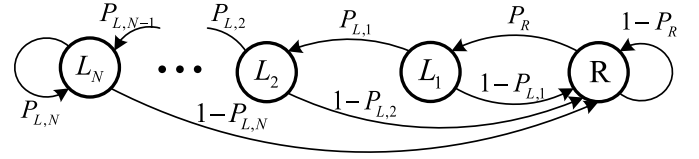


Fig. 7. Loss run-length model.

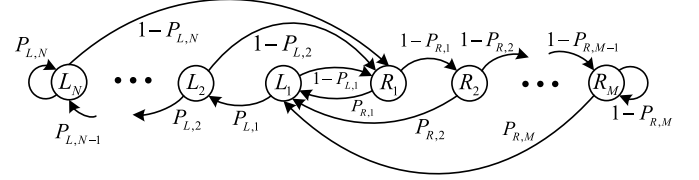


Fig. 8. Receive-loss run-length model.

Finally, [16] shows that the transition probabilities of 4SM can be expressed as

$$P_{13} = \frac{P_{\text{loss}} - P_{\text{iso}}}{E_B (1 - P_{\text{iso}}) (\rho_{\text{loss}} - P_{\text{loss}})},$$

$$P_{31} = \frac{1}{E_B \cdot \rho_{\text{loss}}},$$

$$P_{23} = \frac{E_B \cdot \rho_{\text{loss}} - 1}{E_B - 1},$$

$$P_{32} = \frac{(\rho_{\text{loss}} - 1) \cdot (E_B \cdot \rho_{\text{loss}} - 1)}{\rho_{\text{loss}} \cdot (1 - E_B)},$$

$$P_{14} = \frac{P_{\text{iso}}}{1 - P_{\text{iso}}}.$$

3) *Loss Run-Length Model*: In [18], Sanneck and Carle proposed an extension of the Gilbert model, called the Loss Run-Length model (LRL), as shown in Fig. 7. The LRL model extends the single loss state of the Gilbert model to N loss states, where state $L_k, k = 1, \dots, N$ corresponds to that k losses have occurred in a row. P_R is the probability of moving from the receiving state (R) to the first packet loss (L_1), where $P_{L,k}, k = 1, \dots, N - 1$ is the probability of transition from L_k to L_{k+1} .

Let $[c_j(1), \dots, c_j(S_j)]$ where $c_j(i) = k$ is the number of subsequences consisting of consecutive j 's ($j = 0, 1$) of length i in $\mathbf{b}$, with S_j be the length of the longest such subsequences. Note that $N \equiv S_1, c_j(S_j) > 0$ while some $c_j(i)$ can be 0 for $i < S_j$. Further define the cumulative count vector $[g_j(1), \dots, g_j(S_j)]$, whose elements are given by

$$g_j(k) = \sum_{i=k}^{S_j} c_j(i), \quad j = 0, 1,$$

which counts the number of runs of symbol j with length of at least k . The transition probability of the LRL model fitting $\mathbf{b}$ can be determined as

$$P_{L,k} = \frac{g_1(k+1)}{g_1(k)},$$

$$P_R = f_B, \quad (1)$$

where f_B is the ratio of the number of consecutive 0's subsequences (of any length) in $\mathbf{b}$ to the length of $\mathbf{b}$.

4) *Receive-Loss Run-Length Model*: Sanneck and Carle also proposed Receive-Loss Run-Length model (RLRL) based on LRL model [18] to more accurately characterize consecutive reception. As shown in Fig. 8, while the loss states of the RLRL model remain identical to those of the LRL model, the single receiving state is now expanded to M possible receiving states. R_k corresponds to the k -th successive reception, where $k = 1, \dots, M$. The transition probability $P_{R,k}$ denotes the probability of experiencing a packet loss (and thus transitioning to L_1) given the current state is R_k . Note that $M \equiv S_1$ and $P_{L,k}$ is calculated in (1). It is clear that $P_{R,k}$ can be calculated similarly as

$$P_{R,k} = \frac{g_0(k+1)}{g_0(k)}.$$

where $g_0(k)$ counts the number of runs of consecutive receptions of length at least k . This provides a symmetric characterization of both loss and reception run-lengths.

IV. MODELING PACKET LOSS USING MAP

The aforementioned models are widely utilized in characterizing packet losses in current networks. However, because of the highly dynamic nature of the LEO satellites, we will show in Section VI, that the aforementioned models may not characterize the end-to-end packet loss behavior of LEO satellite networks very well. In this section, we propose to use the MAP to model the packet losses of LEO networks.

A. Introduction to MAP

MAP is an arrival process in which the arrival rate depends on an underlying continuous-time Markov chain (CTMC) with m states [31]. Consider initial probabilities $\beta_0 = [\beta_0(1), \beta_0(2), \dots, \beta_0(m)]$ where $\beta_0(i)$ is that the CTMC starts in state $i = 1, \dots, m$. For CTMC, the time spent in each of the states follows an exponential distribution, whose rate is denoted by λ_i . Arrivals, which correspond to the loss events in our scenario, may only occur at state transitions, either a self-transition or between states. Let p_{ij} and q_{ij} denote the probabilities that jump from state i to j without and with an arrival, respectively. An m -order MAP is characterized by two $m \times m$ matrices $\{\mathbf{D}_0, \mathbf{D}_1\}$, where $\mathbf{D}_0$ and $\mathbf{D}_1$ characterize the hidden (i.e., with no arrival) and the observable transitions (i.e., with arrivals), respectively. They are expressed as

$$\mathbf{D}_0 = \begin{pmatrix} -\lambda_1 & \lambda_1 p_{12} & \cdots & \lambda_1 p_{1m} \\ \lambda_2 p_{21} & -\lambda_2 & \cdots & \lambda_2 p_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_m p_{m1} & \lambda_m p_{m2} & \cdots & -\lambda_m \end{pmatrix},$$

$$\mathbf{D}_1 = \begin{pmatrix} \lambda_1 q_{11} & \lambda_1 q_{12} & \cdots & \lambda_1 q_{1m} \\ \lambda_2 q_{21} & \lambda_2 q_{22} & \cdots & \lambda_2 q_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_m q_{m1} & \lambda_m q_{m2} & \cdots & \lambda_m q_{mm} \end{pmatrix},$$

where $\mathbf{D}_0(i, i) = -\lambda_i \leq 0$, and all the other elements of $\mathbf{D}_0$ and $\mathbf{D}_1$ are nonnegative. For any state, we have $\sum_{j=1, j \neq i}^m \lambda_i p_{ij} + \sum_{j=1}^m \lambda_i q_{ij} = \lambda_i, i = 1, \dots, m$. The generator of the underlying CTMC is given by [37]

$$\mathbf{Q} = \mathbf{D}_0 + \mathbf{D}_1.$$

In this paper, we consider the stationary MAP where the underlying CTMC is irreducible, and assume that β_0 is equal to the stationary probabilities of the CTMC, π , which satisfies

$$\pi \mathbf{Q} = \mathbf{0}, \quad \pi \mathbf{e}^\top = 1, \quad (2)$$

where $\mathbf{e}$ is a row vector with all elements being 1. Note that the average arrival rate of an MAP is

$$\lambda = \pi \mathbf{D}_1 \mathbf{e}^\top. \quad (3)$$

B. Algorithms for MAP Modeling

The task of MAP-based modeling is to estimate the parameters, that is, the elements of the matrices $\mathbf{D}_0, \mathbf{D}_1$ from the loss trace, i.e., $\mathbf{t}$ or equivalently $\mathbf{y}$. For brevity, we use $\phi = \{[\mathbf{D}_0]_{i,j}, [\mathbf{D}_1]_{i,j}, i, j = 1, \dots, m\}$ to refer to these parameters. Note that the observed trace is governed by the CTMC $\{X(t)\}$, which corresponds to a sequence of hidden state transitions, denoted by $\mathbf{x}$. With only incomplete observed data $\mathbf{y}$, ϕ can be estimated using the expectation-maximization (EM) algorithm, in which the main idea is to iteratively update the estimation of ϕ by maximizing the expectation of the complete likelihood function of the process $\mathcal{L}(\phi, \mathbf{x}, \mathbf{y})$.

For the MAP, the $m \times m$ matrix

$$\mathbf{f}(t) = e^{\mathbf{D}_0 t} \cdot \mathbf{D}_1$$

is the transition density matrix where the (i, j) -th element is the state transition density from i to j at time t with an arrival. Accordingly, the matrix

$$\mathbf{F}^c(t) = e^{\mathbf{D}_0 t}$$

contains the complementary cumulative probabilities, where the (i, j) -th element is the probability that the CTMC is in state i and then resides in state j at time t without an arrival.

The derivation of the EM algorithm shows that the estimation of ϕ can be iteratively updated based on the following statistics [38]. Let m_{ij} denote the number of transitions of $\{X(t)\}$ from state i to j in $t \in [0, T]$, T_i be the total time $\{X(t)\}$ spent in state i in $[0, T]$, and a_{ij} be the number of transitions from state i to j with arrival in $[0, T]$. Given a previous round ϕ estimation, the new estimates are

$$[\hat{\mathbf{D}}_0]_{i,j} = \frac{\hat{m}_{ij}}{\hat{T}_i}, \quad i \neq j; i, j = 1, \dots, m, \quad (4)$$

$$[\hat{\mathbf{D}}_1]_{i,j} = \frac{\hat{a}_{ij}}{\hat{T}_i}, \quad i, j = 1, \dots, m. \quad (5)$$

In [30], the statistics for MAP are derived to be

$$\hat{m}_{ij} = \frac{[\mathbf{D}_0]_{i,j}}{\mathcal{L}(\phi, \mathbf{y})} \cdot \sum_{k=1}^{n_t} \mathbf{L}(k-1) \cdot \text{Int}(i, j, k) \cdot \mathbf{R}(k+1), \quad (6)$$

$$\hat{T}_i = \frac{1}{\mathcal{L}(\phi, \mathbf{y})} \cdot \sum_{k=1}^{n_t} \mathbf{L}(k-1) \cdot \text{Int}(i, i, k) \cdot \mathbf{R}(k+1), \quad (7)$$

$$\hat{a}_{ij} = \frac{[\mathbf{D}_1]_{i,j}}{\mathcal{L}(\phi, \mathbf{y})} \cdot \sum_{k=1}^{n_t} \mathbf{L}(k-1) \cdot \mathbf{F}^c(y_k) \cdot \mathbf{1}_i \cdot \mathbf{1}_j^\top \cdot \mathbf{R}(k+1), \quad (8)$$

where

$$\begin{aligned} \text{Int}(i, j, k) &\triangleq \int_{t_{k-1}}^{t_k^-} \mathbf{F}^c(t - t_{k-1}) \cdot \mathbf{1}_i \cdot \mathbf{1}_j^\top \cdot \mathbf{f}(t_k - t) dt \\ &= \int_0^{y_k} \mathbf{F}^c(y) \cdot \mathbf{1}_i \cdot \mathbf{1}_j^\top \cdot \mathbf{f}(y_k - y) dy, \end{aligned} \quad (9)$$

$$\mathbf{L}(k) \triangleq \mathbf{L}(k-1) \cdot \mathbf{f}(y_k), \quad \mathbf{R}(k) \triangleq \mathbf{f}(y_k) \cdot \mathbf{R}(k+1),$$

with $\mathbf{L}(0) = \boldsymbol{\pi}$ and $\mathbf{R}(n_t + 1) = \mathbf{e}^\top$. $\mathbf{L}(k)$ is row vector and $\mathbf{R}(k)$ is column vector. $\mathbf{1}_i$ is a column vector with 0's except for the i -th element being 1. $\mathcal{L}(\phi, \mathbf{y})$ is the likelihood function of the observation and clearly

$$\mathcal{L}(\phi, \mathbf{y}) = \boldsymbol{\pi} \cdot \prod_{k=1}^{n_t} \mathbf{f}(y_k) \cdot \mathbf{e}^\top.$$

Unfortunately, computational difficulties often arise in computing (6), (7) and (8), which involve the multiplication of possibly very small probabilities and integrals of matrix exponential (9). In [30], a scaling procedure is claimed to stabilize the computation. However, no details were specified. The uniformization technique is used to approximate the integrals by a truncated summation of the matrix powers. This approach, however, requires to adaptively choose truncation points and often results in heavy for-loop summations, which is difficult to be efficiently implemented in practice. To overcome these difficulties, we use the scaling procedure of [39] along with the log-sum-exp trick to achieve numerical stability. In addition, the matrix diagonalization approach of [38] is applied for the integrals of matrix exponential. These techniques have been used for fitting the Markov-modulated Poisson process (MMPP).² We show that they apply to MAP too.

The key to scaling is to note that the likelihood

$$\mathcal{L}(\phi, \mathbf{y}) = \prod_{l=1}^{n_t} \Pr\{y_l | \mathbf{y}^{l-1}\},$$

where $\Pr\{y_l | \mathbf{y}^{l-1}\}$ is the conditional probability of observing y_l given the previously observed $(l-1)$ -length sequence. Let $c_k = \Pr\{y_k | \mathbf{y}^{k-1}\}$. We redefine scaled $\mathbf{L}(k)$ and $\mathbf{R}(k)$ as

$$\mathbf{L}_s(k) \triangleq \boldsymbol{\pi} \cdot \prod_{l=1}^k \frac{\mathbf{f}(y_l)}{c_l}, \quad k = 1, \dots, n_t,$$

$$\mathbf{R}_s(k) \triangleq \boldsymbol{\pi} \cdot \prod_{l=k}^{n_t} \frac{\mathbf{f}(y_l)}{c_l} \cdot \mathbf{e}^\top, \quad k = n_t, \dots, 1.$$

With this definition, we can verify that

$$c_k = \mathbf{L}_s(k-1) \cdot \mathbf{f}(y_k) \cdot \mathbf{e}^\top. \quad (10)$$

This leads to scaled recursions of $\mathbf{L}_s(k)$ and $\mathbf{R}_s(k)$ as

$$\mathbf{L}_s(k) = \frac{\mathbf{L}_s(k-1) \mathbf{f}(y_k)}{\mathbf{L}_s(k-1) \cdot \mathbf{f}(y_k) \cdot \mathbf{e}^\top}, \quad (11)$$

$$\mathbf{R}_s(k) = \frac{\mathbf{f}(y_k) \cdot \mathbf{R}_s(k+1)}{\mathbf{L}_s(k-1) \cdot \mathbf{f}(y_k) \cdot \mathbf{e}^\top}. \quad (12)$$

²MMPP can be viewed as a special form of MAP with diagonal $\mathbf{D}_1$.

Thus, the computation of the statistics can be re-written as

$$\hat{m}_{ij} = \sum_{k=1}^{n_t} \frac{[\mathbf{D}_0]_{i,j}}{c_k} \mathbf{L}_s(k-1) \cdot \text{Int}(i, j, k) \cdot \mathbf{R}_s(k+1), \quad (13)$$

$$\hat{T}_i = \sum_{k=1}^{n_t} \frac{1}{c_k} \cdot \mathbf{L}_s(k-1) \cdot \text{Int}(i, i, k) \cdot \mathbf{R}_s(k+1), \quad (14)$$

$$\hat{a}_{ij} = \sum_{k=1}^{n_t} \frac{[\mathbf{D}_1]_{i,j}}{c_k} \cdot \mathbf{L}_s(k-1) \cdot \mathbf{F}^c(y_k) \cdot \mathbf{1}_i \cdot \mathbf{1}_j^\top \cdot \mathbf{R}_s(k+1). \quad (15)$$

Since $\mathbf{L}_s(k)$, $\mathbf{R}_s(k)$, c_k can become very small as k increases, most of the above computation should be performed in the log domain and only be converted back when summing in (13)-(15). Taking c_k computation as an example,

$$\begin{aligned} \log c_k &= \log \sum_{i_0=1}^m \sum_{j_0=1}^m [\mathbf{L}_s(k-1)]_{i_0} [\mathbf{f}(y_k)]_{i_0, j_0} \\ &= \log \sum_{i_0=1}^m \sum_{j_0=1}^m e^{\log[\mathbf{L}_s(k-1)]_{i_0} + \log[\mathbf{f}(y_k)]_{i_0, j_0}} \\ &= \text{LSE} \{ \log[\mathbf{L}_s(k-1)]_{i_0} + \log[\mathbf{f}(y_k)]_{i_0, j_0} \} \end{aligned}$$

where $\text{LSE}\{\cdot\}$ is the log-sum-exp of all the elements, i.e., for all $i_0, j_0 = 1, \dots, m$.

To compute (9), assume that $\mathbf{D}_0$ can be diagonalized as $\mathbf{D}_0 = \mathbf{V} \boldsymbol{\Lambda} \mathbf{V}^{-1}$, where $\boldsymbol{\Lambda}$ contains eigenvalues $\nu_1, \dots, \nu_m$. The matrix exponential

$$\begin{aligned} e^{\mathbf{D}_0 t} &= \sum_{n=0}^{\infty} \frac{1}{n!} (\mathbf{D}_0 t)^n = \sum_{n=0}^{\infty} \frac{t^n}{n!} (\mathbf{V} \boldsymbol{\Lambda} \mathbf{V}^{-1})^n \\ &= \mathbf{V} \left(\sum_{n=0}^{\infty} \frac{t^n}{n!} \boldsymbol{\Lambda}^n \right) \mathbf{V}^{-1} = \mathbf{V} e^{\boldsymbol{\Lambda} t} \mathbf{V}^{-1}. \end{aligned}$$

Therefore, the integral $\text{Int}(i, j, k)$ transforms to

$$\begin{aligned} &\int_0^{y_k} \mathbf{V} e^{\boldsymbol{\Lambda} y} \mathbf{V}^{-1} \cdot \mathbf{1}_i \cdot \mathbf{1}_j^\top \cdot \mathbf{V} e^{\boldsymbol{\Lambda} (y_k - y)} \mathbf{V}^{-1} \mathbf{D}_1 dy \\ &= \mathbf{V} \left(\int_0^{y_k} e^{\boldsymbol{\Lambda} y} \cdot (\mathbf{u}_i \mathbf{v}_j^\top) \cdot e^{\boldsymbol{\Lambda} (y_k - y)} dy \right) \mathbf{V}^{-1} \mathbf{D}_1, \end{aligned}$$

where $\mathbf{u}_i$ and $\mathbf{v}_j^\top$ are the i -th column of j -th row of $\mathbf{V}^{-1}$ and $\mathbf{V}$, respectively. Since $\boldsymbol{\Lambda}$ is diagonal, the integral of the matrix exponential can be reduced to element-wise scalar integrals. The (i_0, j_0) -th element of the matrix corresponding to the integral in the parenthesis, $i_0, j_0 = 1, \dots, m$, is

$$\begin{aligned} &[\mathbf{u}_i]_{i_0} \cdot [\mathbf{v}_j^\top]_{j_0} \cdot \int_0^{y_k} e^{\nu_{i_0} y} \cdot e^{\nu_{j_0} (y_k - y)} dy \\ &= \begin{cases} \frac{e^{\nu_{i_0} y_k} - e^{\nu_{j_0} y_k}}{\nu_{i_0} - \nu_{j_0}}, & \nu_{i_0} \neq \nu_{j_0}, \\ y_k e^{\nu_{i_0} y_k}, & \nu_{i_0} = \nu_{j_0}. \end{cases} \end{aligned}$$

Overall, we obtain Algorithm 1 for fitting the MAP model from the loss trace $\mathbf{y}$.

Algorithm 1 EM-Based MAP Fitting

Data: $\mathbf{y} = \{y_k : 1 \leq k \leq n_t\}$ and MAP order m .
Result: $\{\mathbf{D}_0, \mathbf{D}_1\}$.

- 1 Initialize max iterations M_i , ToL $\leftarrow 0.01$, step $\leftarrow 1$.
- 2 Determine $\mathbf{D}_0, \mathbf{D}_1$ according to random initialization.
- 3 $\mathbf{Q} \leftarrow \mathbf{D}_0 + \mathbf{D}_1$.
- 4 Solve $\pi \mathbf{Q} = \mathbf{0}$ subject to $\pi \mathbf{e}^\top = 1$ to obtain π .
- 5 **while** ψ_0 or $\psi_1 > \text{ToL}$ & step $\leq M_i$ **do**
- 6 **E-Step:**
- 7 Let $\mathbf{L}_s(0) \leftarrow \pi$, $\mathbf{R}_s(n_t + 1) \leftarrow \mathbf{e}^\top$.
- 8 **for** $k = 1, \dots, n_t$ **do**
- 9 | Compute (10), (11), (12) as $c_k, \mathbf{L}_s(k), \mathbf{R}_s(k)$.
- 10 **end**
- 11 **for** $i, j = 1, \dots, m$ **do**
- 12 | Compute (13), (14), (15) as $\hat{m}_{ij}, \hat{T}_i, \hat{a}_{ij}$.
- 13 **end**
- 14 $\mathcal{L}(\phi, \mathbf{y}) \leftarrow \pi \cdot \mathbf{R}_s(1)$.
- 15 **M-Step:**
- 16 $[\hat{\pi}]_i \leftarrow [\pi]_i \cdot \mathbf{1}_i \cdot \mathbf{R}_s(1) / \mathcal{L}(\phi, \mathbf{y})$, for $i = 1, \dots, m$.
- 17 **for** $i, j = 1 : m$ **do**
- 18 | Compute (4), (5) as $[\hat{\mathbf{D}}_0]_{ij}, [\hat{\mathbf{D}}_1]_{ij}$.
- 19 | $[\hat{\mathbf{D}}_0]_{i,i} \leftarrow -\left(\sum_{j \neq i} [\hat{\mathbf{D}}_0]_{i,j} + \sum_j [\hat{\mathbf{D}}_1]_{i,j}\right)$.
- 20 **end**
- 21 $\psi_0 \leftarrow \max \left| \left([\hat{\mathbf{D}}_0]_{ij} - [\mathbf{D}_0]_{ij} \right) / [\mathbf{D}_0]_{ij} \right|$,
- 22 $\psi_1 \leftarrow \max \left| \left([\hat{\mathbf{D}}_1]_{ij} - [\mathbf{D}_1]_{ij} \right) / [\mathbf{D}_1]_{ij} \right|$.
- 23 $\pi \leftarrow \hat{\pi}, \mathbf{D}_0 \leftarrow \hat{\mathbf{D}}_0, \mathbf{D}_1 \leftarrow \hat{\mathbf{D}}_1$.
- 24 step $\leftarrow \text{step} + 1$.
- 25 **end**
- 26 **return** $\{\mathbf{D}_0, \mathbf{D}_1\}$.

V. MAP/D/1 QUEUE AND FEC DECODING DELAY

In the previous section, we have proposed MAP to model LEO satellite networks' packet loss behavior. In this section, under this modeling, we analyze the FEC decoding delay based on the busy period analysis of a MAP/D/1 queue.

A. Introduction to MAP/D/1 Queue

Before proceeding, a brief introduction to the MAP/D/1 queue and its analysis is presented first. An MAP/D/1 queue corresponds to a single-server system with arrivals governed by an MAP specified by the $m \times m$ matrices $\{\mathbf{D}_0, \mathbf{D}_1\}$, with deterministic service times of constant duration $1/\mu$. Let $N(t)$ be the number of arrivals in $[0, t)$ and $J(t)$ be the phase at time t . A key quantity for analyzing the MAP/D/1 queue is the matrix generating function

$$\mathbf{D}(z) \triangleq \sum_{n=0}^{\infty} \mathbf{D}_n z^n = \mathbf{D}_0 + z\mathbf{D}_1, \quad |z| \leq 1.$$

Define the matrix $\mathbf{P}(n, t)$ whose (i, j) -th element is

$$\mathbf{P}_{ij}(n, t) \triangleq \mathbf{P}(N(t) = n, J(t) = j \mid N(0) = 0, J(0) = i).$$

$\mathbf{P}(n, t)$ characterizes the probabilities of n arrivals within $[0, t)$ given that the underlying CTMC transitions from phase i to j . Using Chapman-Kolmogorov equations, we have [32]

$$\mathbf{P}'(n, t) = \sum_{r=0}^n \mathbf{P}(r, t) \mathbf{D}_{n-r}, \quad n \geq 0, t \geq 0,$$

$$\mathbf{P}(0, 0) = \mathbf{I},$$

where $\mathbf{I}$ denotes the $m \times m$ unit matrix. The matrix-generating function $\mathbf{P}^*(z, t)$ is defined by

$$\mathbf{P}^*(z, t) \triangleq \sum_{n=0}^{\infty} \mathbf{P}(n, t) z^n, \quad \text{for } |z| \leq 1, t \geq 0. \quad (16)$$

We can verify that the Chapman-Kolmogorov equations yield

$$\frac{\partial}{\partial t} \mathbf{P}^*(z, t) = \mathbf{P}^*(z, t) \mathbf{D}(z), \quad |z| \leq 1,$$

with $\mathbf{P}^*(z, 0) = \mathbf{I}$. Therefore, we obtain

$$\mathbf{P}^*(z, t) = e^{\mathbf{D}(z)t}, \quad \text{for } |z| \leq 1, t \geq 0. \quad (17)$$

The key to analyzing the MAP/D/1 queue is to study the embedded Markov renewal process (MRP) embedded at departure epochs. Define $\zeta(t)$ as the numbers of customers in the system (including that in service) at time t . Let ϑ_τ as the time of the τ -th departure (service completion), with $\vartheta_0 = 0$. The interval $\vartheta_{\tau+1} - \vartheta_\tau$ corresponds to the duration of the $(\tau + 1)$ -th service. The triplet $\{\zeta(\vartheta_\tau), J(\vartheta_\tau), \vartheta_{\tau+1} - \vartheta_\tau\}$ for $\tau \geq 0$ then forms a semi-Markov process on the state space $\Omega = \{(i, j) : i \geq 0, 1 \leq j \leq m\}$, i.e., i customers in the queue while the underlying CTMC is at phase j . For convenience, we define level $\mathbf{i} = \{(i, 1), \dots, (i, m)\}$, which contains all the states of the same i but different j .

Define matrix $\hat{\mathbf{A}}_n(t)$ where the elements $\hat{a}_{i,j}(n, t), 1 \leq i, j \leq m$ are the conditional probability that the last departure at time 0 left the queue busy with the underlying CTMC in phase i and the next departure, which occurs no later than time t , sees the CTMC in phase j and there were n arrivals during the service of the departed customer. The matrix can be expressed as a function of $\mathbf{P}(n, t)$ and the service time distribution $F(\cdot)$, which is

$$\hat{\mathbf{A}}_n(t) = \int_0^t \mathbf{P}(n, u) dF(u), \quad n \geq 0, t \geq 0. \quad (18)$$

The stationary $\hat{\mathbf{A}}_n(t)$ is

$$\mathbf{A}_n \triangleq \hat{\mathbf{A}}_n(\infty) = \int_0^\infty \mathbf{P}(n, t) dF(t) = \mathbf{P}(n, d),$$

where the last equality is due to constant service time $d = 1/\mu$. Define

$$\mathbf{A}^*(z, s) \triangleq \sum_{n=0}^{\infty} z^n \int_0^\infty e^{-st} d\hat{\mathbf{A}}_n(t), \quad \text{Re}(s) \geq 0, |z| \leq 1.$$

Using (18), we obtain

$$\begin{aligned} \mathbf{A}^*(z, s) &= \int_0^\infty e^{-st} \sum_{n=0}^{\infty} z^n \mathbf{P}(n, t) dF(t) \\ &= \int_0^\infty e^{-(\mathbf{I} - \mathbf{D}(z))t} dF(t), \end{aligned}$$

where the second equation comes from (16) and (17). The expected number of arrivals during a service that starts when the underlying CTMC is in phase j , denoted by the j -th element of a vector β , can then be determined as

$$\begin{aligned}\beta^\top &= \sum_{k=1}^{\infty} k \mathbf{A}_k \mathbf{e}^\top \\ &= \frac{d}{dz} \mathbf{A}^*(z, 0) \Big|_{z=1} \mathbf{e}^\top \\ &= \frac{\lambda}{\mu} \mathbf{e}^\top + (\mathbf{I} - \mathbf{A})(\mathbf{e}^\top \boldsymbol{\pi} - \mathbf{Q})^{-1} \mathbf{D}_1 \mathbf{e}^\top,\end{aligned}$$

with

$$\mathbf{A} \triangleq \mathbf{A}^*(1, 0) = \int_0^\infty e^{\mathbf{D}(1)t} dF(t) = \int_0^\infty e^{\mathbf{Q}t} dF(t) = e^{\mathbf{Q}d},$$

where again the last equality is due to constant service time.

Define matrix $\mathbf{G}(n, t)$, whose the (j, j') -th elements $g_{j,j'}(n, t)$, $1 \leq j, j' \leq m, n \geq 1, t \geq 0$, corresponds to the condition probability that the first passage reaches state (i, j') from state $(i+1, j)$ in exactly n transitions within time t . Define the transform of $\mathbf{G}(n, t)$

$$\mathbf{G}^*(z, s) \triangleq \sum_{n=1}^{\infty} z^n \int_0^\infty e^{-st} d\mathbf{G}(n, t), \quad |z| \leq 1, \text{Re}(s) \geq 0.$$

A fundamental equation that connects the arrival process with $\mathbf{G}(n, t)$, as proved in [40], is

$$\mathbf{G}^*(z, s) = z \sum_{n=0}^{\infty} \mathbf{A}_n(s) \mathbf{G}^{*n}(z, s),$$

where $\mathbf{A}_n(s) \triangleq \int_0^\infty e^{-st} d\hat{\mathbf{A}}_n(t)$. This equation plays a central role in characterizing MAP/D/1's busy period, and is essentially an analogue version of the well-known Takacs's equation for the M/G/1's busy period. Denote

$$\mathbf{G} \triangleq \mathbf{G}^*(1, 0) = \sum_{n=0}^{\infty} \mathbf{A}_n \mathbf{G}^n.$$

and the vector $\mathbf{g}$ being the solution of

$$\mathbf{g} \mathbf{G} = \mathbf{g}, \quad \mathbf{g} \mathbf{e}^\top = 1.$$

$\mathbf{G}$ is stochastic when $\rho < 1$. $\mathbf{G}$ and $\mathbf{g}$ are keys to compute the moments of the busy period.

B. Analysis of FEC Decoding Delay

Note that the key to characterizing the FEC delay lies in the analysis of the duration of the busy period of the decoder. A busy period begins from the expected reception time of $\mathbf{s}_{i_{\text{ord}}+1}$ if it were not lost (although the decoder is activated when seeing an out-of-order packet, the decoding process should have virtually started before that). Each subsequent received repair packet provides a linearly independent row in the decoding matrix $\mathbf{A}_c$ if a sufficiently large $\mathbb{F}_q$ is used for encoding. The busy period ends when the number of repaired packets received is equal to the number of source packets lost during the busy period. This process may be considered as an analog to serving customers in a virtual queue if we assume that the repair sending process is independent of the packet

loss process. We treat the source packet that triggered the busy period of the decoder as a customer arriving at the empty virtual queue [33], and treat all the following lost packets before in-order delivery resumes as following customers. The sending of each repair packet is considered a service, and the insertion interval is the service time. When the number of services equals to the number of arrived customers, the queue becomes empty again. Thus, the duration of the decoder's busy period, denoted by B , can be approximated by the busy period of the queue. Formally, we have the following proposition when the loss process is an MAP.

Proposition 1: For an FEC scheme sending repair packets with fixed interval d over a connection where packet loss follows an MAP characterized by m -dimension matrices $\{\mathbf{D}_0, \mathbf{D}_1\}$ with average loss rate $\lambda < \mu \triangleq 1/(d \cdot R_c)$, the expected busy period of the decoder, $E\{B\}$, is approximately equal to the expected busy period of the MAP/D/1 queue with the same arrival process and constant service rate μ , denoted by μ_{BP} . In addition,

$$\begin{aligned}\mu_{BP} &= \frac{1}{\mu} \mathbf{g}(-\mathbf{D}_0)^{-1} \mathbf{D}_1 \\ &\quad \times (\mathbf{I} - \mathbf{G} + \mathbf{e}^\top \mathbf{g}) [\mathbf{I} - \mathbf{A} + (\mathbf{e}^\top - \beta^\top) \mathbf{g}]^{-1} \mathbf{e}^\top,\end{aligned}\quad (19)$$

which can be calculated using the numerical recipe listed in Algorithm 2.³

The readers are referred to [32, Section 4.2] for the detailed derivation of (19). To numerically evaluate μ_{BP} , the computation should start with the calculation of $\mathbf{P}(n, t)$. This can be efficiently achieved using the method proposed in [41], outlined as follows. Define matrix $\mathbf{K}$ that describes each step of the transfer as

$$\mathbf{K} \triangleq \begin{bmatrix} \mathbf{E}_0 & \mathbf{E}_1 & \cdots \\ & \mathbf{E}_0 & \mathbf{E}_1 & \cdots \\ & & \ddots & \ddots \end{bmatrix}, \quad \mathbf{E}_0 \triangleq \mathbf{I} + \frac{1}{\theta} \mathbf{D}_0, \quad \mathbf{E}_1 \triangleq \frac{1}{\theta} \mathbf{D}_1,$$

where $\theta = \max_i |(\mathbf{D}_0)_{i,i}|$ is the maximum absolute value of the diagonal elements of the $\mathbf{D}_0$ matrix. To compute $\mathbf{P}(n, t)$, we define sub-matrix $\mathbf{F}_{n,j}$. With respect to $j \geq 1$, $\mathbf{F}_{n,j}$ satisfies

$$\mathbf{F}_{n,j} = \mathbf{E}_0 \mathbf{F}_{n,j-1} + \mathbf{E}_1 \mathbf{F}_{n-1,j}.$$

We obtain $\mathbf{P}(n, t)$ as follows [31]

$$\mathbf{P}(n, t) = \sum_{j=n}^{\infty} e^{-\theta t} \frac{(\theta t)^j}{j!} \mathbf{F}_{n,j}.$$

In practical calculations, infinite summation requires truncation. In Algorithm 2, we have adopted this technique at several places. For example, for the computation of $\mathbf{P}(n, t)$ we need to choose n^* such that

$$\sum_{j=n^*+1}^{\infty} e^{-\theta t} \frac{(\theta t)^j}{j!} < \epsilon,$$

where ϵ is a very small positive number indicating a preset accuracy threshold [41].

³The numerical scripts for Algorithms 1 and 2 are available upon requests.

Algorithm 2 MAP/D/1 Busy Period Calculation

Input: $\{\mathbf{D}_0, \mathbf{D}_1\}$, service rate μ , small $\epsilon > 0$.
Output: μ_{BP} .

- 1 Generator $\mathbf{Q} \leftarrow \mathbf{D}_0 + \mathbf{D}_1$.
- 2 Calculate $\boldsymbol{\pi}$ of $\mathbf{Q}$ using Equation (2).
- 3 Arrival rate $\lambda \leftarrow \boldsymbol{\pi} \mathbf{D}_1 \mathbf{e}$.
- 4 $\mathbf{A} \leftarrow e^{\mathbf{Q}/\mu}$.
- 5 $\boldsymbol{\beta}^\top \leftarrow \frac{\lambda}{\mu} \mathbf{e}^\top + (\mathbf{I} - \mathbf{A})(\mathbf{e}^\top \boldsymbol{\pi} - \mathbf{Q})^{-1} \mathbf{D}_1 \mathbf{e}^\top$.
- 6 $\theta \leftarrow \max_i |(\mathbf{D}_0)_{i,i}|$.
- 7 Normalize transfer matrices $\mathbf{E}_0 \leftarrow \frac{\mathbf{D}_0}{\theta} + \mathbf{I}, \mathbf{E}_1 \leftarrow \frac{\mathbf{D}_1}{\theta}$.
- 8 Calculate $\gamma_j \leftarrow e^{-\theta/\mu} \frac{(\theta/\mu)^j}{j!}, 0 \leq j \leq n^*$, such that $\sum_{j=n^*+1}^{\infty} \gamma_j < \epsilon$.
- 9 $\gamma_j \leftarrow 0, j > n^*$.
- 10 $\mathbf{F}_{0,1} \leftarrow \mathbf{I}$.
- 11 **for** $j = 2 : 1 : n^*$ **do**
- 12 $\mathbf{F}_{0,j} \leftarrow \mathbf{E}_0 \mathbf{F}_{0,j-1}$.
- 13 **end**
- 14 $\mathbf{P}(0, 1/\mu) \leftarrow e^{\mathbf{D}_0/\mu}, \mathbf{P}_{\text{tot}} \leftarrow \mathbf{P}(0, 1/\mu)$.
- 15 $\mathbf{A}_0 \leftarrow \mathbf{P}(0, 1/\mu), n \leftarrow 1$.
- 16 **while** $\min_i \left(\sum_{j=1}^m [\mathbf{P}_{\text{tot}}]_{i,j} \right) < 1 - \epsilon$ **do**
- 17 $\mathbf{F}_{n,1} \leftarrow \mathbf{E}_1 \mathbf{F}_{n-1,1}$.
- 18 $\mathbf{P}(n, 1/\mu) \leftarrow \gamma_n \mathbf{F}_{n,1}$.
- 19 **for** $j = 2 : 1 : n^*$ **do**
- 20 $\mathbf{F}_{n,j} \leftarrow \mathbf{E}_0 \mathbf{F}_{n,j-1} + \mathbf{E}_1 \mathbf{F}_{n-1,j}$.
- 21 $\gamma_n \leftarrow \gamma_n \cdot \frac{\theta/\mu}{n-1+j}$.
- 22 $\mathbf{P}(n, 1/\mu) \leftarrow \mathbf{P}(n, 1/\mu) + \gamma_n \mathbf{F}_{n,j}$.
- 23 **end**
- 24 $\mathbf{P}_{\text{tot}} \leftarrow \mathbf{P}_{\text{tot}} + \mathbf{P}(n, 1/\mu)$.
- 25 $\mathbf{A}_n \leftarrow \mathbf{P}(n, 1/\mu)$.
- 26 $n \leftarrow n + 1$.
- 27 **end**
- 28 $n_1^* \leftarrow n$.
- 29 Use steps (30)-(39) to compute $\mathbf{G} = \sum_{n=0}^{n_1^*} \mathbf{A}_n \mathbf{G}^n$.
- 30 Initialize $\chi \leftarrow 1, \mathbf{G} \leftarrow \mathbf{A}_0$.
- 31 **while** $\chi > \epsilon$ **do**
- 32 $\mathbf{G}_{\text{old}} \leftarrow \mathbf{G}, \mathbf{T} \leftarrow \mathbf{G}$.
- 33 $\mathbf{G} \leftarrow \mathbf{A}_0 + \mathbf{A}_1 \mathbf{G}$.
- 34 **for** $2 \leq n \leq n_1^* - 2$ **do**
- 35 $\mathbf{T} \leftarrow \mathbf{T} \mathbf{G}_{\text{old}}$.
- 36 $\mathbf{G} \leftarrow \mathbf{G} + \mathbf{A}_n \mathbf{T}$.
- 37 **end**
- 38 $\chi \leftarrow \|\mathbf{G} - \mathbf{G}_{\text{old}}\|_{\infty}$.
- 39 **end**
- 40 Solve $\mathbf{g} \mathbf{G} = \mathbf{g}$ subject to $\mathbf{g} \mathbf{e}^\top = 1$ to obtain $\mathbf{g}$.
- 41 Calculate μ_{BP} using Equation (19).

We also need to choose n_1^* for the computation of the infinite matrix sequence $\{\mathbf{A}_n\}$, such that [31]

$$\left\| \sum_{n=n_1^*+1}^{\infty} \mathbf{A}_n \mathbf{e}^\top - \mathbf{e}^\top \right\|_{\infty} < \epsilon.$$

In Algorithm 2, the matrix $\mathbf{P}_{\text{tot}}$ accumulates all possible state transition probabilities within the time interval d , approximating the total transition probability. By computing the row sums

of $\mathbf{P}_{\text{tot}}$, the truncation point is dynamically adjusted to ensure that the total transition probability from the current state to all possible states approaches 1, covering all possible events.

VI. PERFORMANCE EVALUATION

In this section, we evaluate the proposed MAP loss model and validate the proposed FEC decoding analysis under the model, based on measured real-life LEO loss traces.

A. Starlink Trace Acquisition

At present, Starlink is one of the main source from which we can obtain the real-life end-to-end data over LEO satellite network. In this study, the measurement system shown in Fig. 1 is deployed in two locations. One has the Starlink UT installed in Victoria, British Columbia, Canada, with a remote server located at the Akamai Linode data center in Seattle, USA, near the closest PoP of the UT; another places the UT in Brühl, Germany, and connects to a remote server in Frankfurt, Germany. Both setups ensure a one-hop bent-pipe LEO connection to avoid interference from other ground segment services to the largest extent.

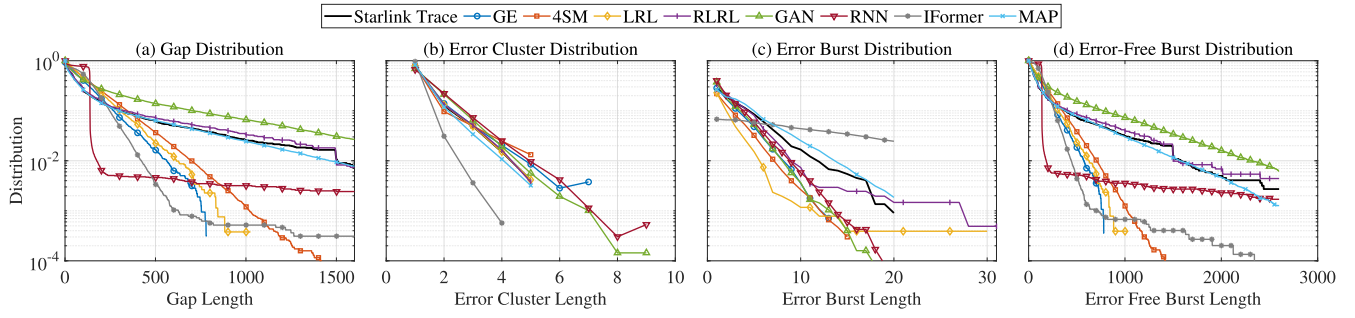
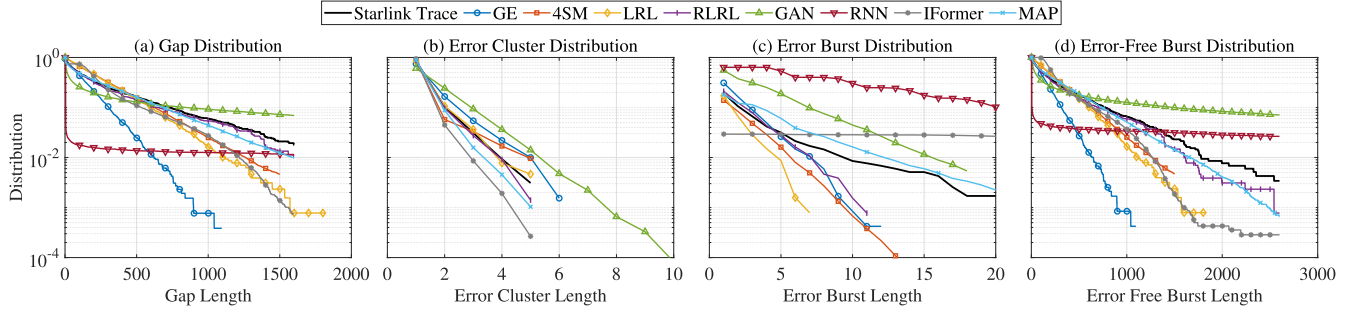
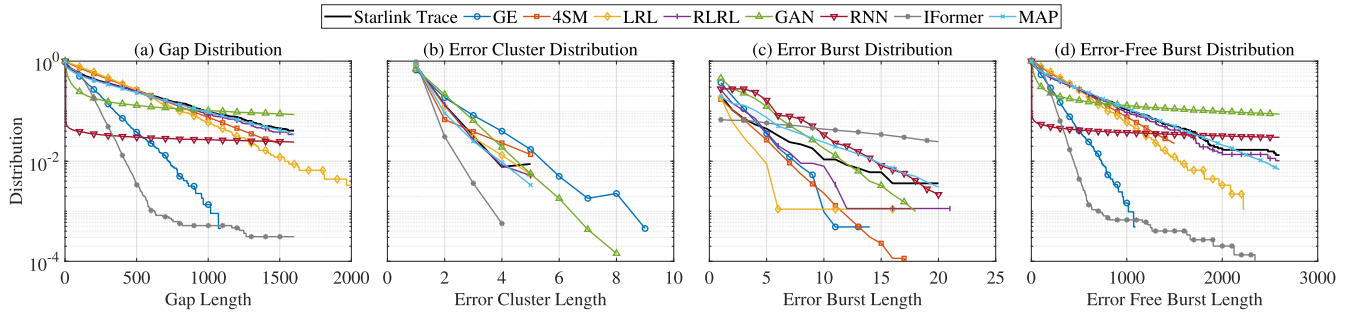
Building upon the experimental setup described in [12], we perform active measurements between a sender (connected to the Starlink Ethernet) and a receiver (on the terrestrial network). We deploy a cronjob service to (re)start `irtt`⁴ for active probing, transmitting UDP packets at a fixed 10 ms interval ($R_c = 100$ packets/second). This sampling rate provides sufficient temporal resolution to characterize satellite link dynamics while maintaining measurement stability. For each probe packet, we record the sequence number, transmission and reception timestamps, and packet loss status, with clock synchronization maintained via NTP. The analysis of packet delivery status, focusing on whether each packet is successfully delivered or lost, enables us to obtain the binary trace sequence $\mathbf{b}$ and the loss interval sequence $\mathbf{y}$.

B. Packet Loss Data Modeling

We present the modeling results for packet loss traces collected across three distinct scenarios designed to evaluate model performance under different locations, weather conditions, and time periods. The data include measurements from Victoria, Canada under clear sky conditions on September 22, 2024 (19:00-20:00 local time), along with two sets of measurements from Brühl, Germany comprising rainy conditions on August 4, 2025 (19:00-20:00 local time) and cloudy conditions on August 9, 2025 (01:00-02:00 local time). This selection includes both peak and off-peak Internet usage periods, with peak hours defined as 15:00 to 22:00 [5].⁵ For the existing Markov chain-based models, the formulas in Section III are applied to obtain the fitted parameters. For MAP, Algorithm 1 is applied to fit the data and obtain the matrices $\{\mathbf{D}_0, \mathbf{D}_1\}$, for which we have tried $m = 16$. This model order is empirically determined to achieve a balance between accuracy and numerical computation costs in capturing packet loss characteristics.

⁴<https://github.com/heistp/irtt>

⁵Readers can find more of our monthly-updated Starlink trace dataset at <https://github.com/clarkzjw/LENS>

Fig. 9. Modeling performance comparison with Starlink trace data from Canada collected under clear sky (September 22, 2024); $\eta = 5$.Fig. 10. Modeling performance comparison with Starlink trace data from Germany collected under rain (August 4, 2025); $\eta = 5$.Fig. 11. Modeling performance comparison with Starlink trace data from Germany under cloudy skies (August 9, 2025); $\eta = 5$.TABLE I
MODELS' RMSE OF FITTING STARLINK LOSS TRACE

Model	Sept. 22, 2024 (Canada)				Aug. 4, 2025 (Germany)				Aug. 9, 2025 (Germany)			
	G	EC	EB	EFB	G	EC	EB	EFB	G	EC	EB	EFB
GE	0.0660	0.0145	0.0234	0.0660	0.1061	0.0552	0.0393	0.1169	0.1622	0.0745	0.0535	0.1789
4SM	0.0681	0.0143	0.0339	0.0543	0.0664	0.0240	0.0169	0.0428	0.0623	0.0321	0.0125	0.0388
LRL	0.0871	0.0035	0.0446	0.0691	0.0722	0.0036	0.0276	0.0575	0.0816	0.0063	0.0227	0.0573
RLRL	0.0082	0.0055	0.0146	0.0066	0.0137	0.0074	0.0106	0.0122	0.0103	0.0032	0.0192	0.0088
GAN	0.0732	0.0625	0.0254	0.0562	0.0955	0.1288	0.1342	0.0799	0.1406	0.0658	0.0845	0.1160
RNN	0.1290	0.0753	0.0281	0.1097	0.2133	0.1512	0.3472	0.1710	0.2522	NaN	0.0826	0.2108
IFormer	0.0777	0.0870	0.0649	0.0847	0.0693	0.0514	0.0450	0.0970	0.1676	0.0938	0.0400	0.1418
MAP	0.0078	0.0155	0.0120	0.0054	0.0151	0.0210	0.0178	0.0143	0.0216	0.0329	0.0221	0.0061

Given the fitted models, we generate random loss traces and then empirically evaluate the metrics of Section II for each of the models. The metrics are compared with existing models in Figs. 10, 11 and 9 for the three dates, respectively. The average root mean square error (RMSE) of fitting for each curve is listed in Table I. In the figures, we have plotted the metrics

for the original data (black curve) for reference. We have also included the metrics for the simulated traces generated by three ML-based models, namely a generative adversarial network (GAN) [28], a long short-term memory (LSTM) network [42] which is a type of recurrent neural networks (RNNs), and a recent Transformer variant, IFormer [43]. The ML models are

TABLE II
NUMBER OF MODEL STATES

Model	GE	4SM	LRL	RLRL	MAP
States	2	4	29 (Sept. 22) 11 (Aug. 4) 20 (Aug. 9)	3430 (Sept. 22) 3265 (Aug. 4) 4740 (Aug. 9)	16

trained based on measurement data collected during two days before (inclusive) the dates of the corresponding figures, which consist of traces of transmitting more than 100 million packets with 60514, 11941, and 24122 loss events, respectively.⁶

The figures show that GE and 4SM models, which are very commonly used in the terrestrial networks, fail to capture almost all the 4 metrics in the LEO scenario. The two models are only close to the Starlink trace for the EC distribution as shown in Fig. 10(b), Fig. 11(b) and Fig. 9(b), where the other models also perform well (and better). In the other three metrics, the fitted models are far from the Starlink data, suggesting that the two models are not suitable for LEO networks. Similarly, the LRL model, although allows for many more states than GE and 4SM models, is not capable of fitting the characteristics of the loss trace. From Fig. 10(a), (d), Fig. 11(a), (d) and Fig. 9(a), (d), it is seen that the RLRL model can fit the traces relatively well. Furthermore, since the RLRL model uses a large number of states, it can even fit the sudden drops of the gap and EFB distributions. However, Fig. 10(c), Fig. 11(c) and Fig. 9(c) shows that it cannot capture the EB distribution of the traces.

The figures show that the three ML-based models do not perform well in general. Among the methods, GAN is much better than the other two in most cases, and is especially close to MAP in Fig. 10. However, Figs. 11(d) and 9(d) show that it fails to capture the EFB distribution for the Germany data, which exhibits fewer losses than Canada's. RNN and IFormer perform poorly, where RNN fails to capture EC at all (Fig. 11(b), 9(b)) and IFormer fails to capture the EB distribution in Figs. 10(c), 11(b) and 9(c). The reason may be multi-fold. First, considering the expense of data collection and computational costs, we have limited the training to only days of data. Due to the highly dynamic nature of the LEO loss pattern, it could make the ML models difficult to learn the hidden pattern. Second, we note that we have used "out-of-box" ML models with minimal customization when fitting the data, and hence the models may not be powerful enough. This is intentional for reproducibility and avoiding overfitting with heavy customization/tuning.

For the four metrics, Fig. 10, Fig. 11 and Fig. 9 show that the MAP model achieves the best performance compared to the other models, and is almost the only model that can be close to the trace for the EB distribution (Figs. 10(c), 11(c) and 9(c)). This suggests that MAP is a feasible model for capturing the packet loss characteristics of LEO networks.

⁶The training scripts are available at <https://github.com/yeliqseu/leo-ml-loss-models>, where the configuration parameters are self-explained to save space here.

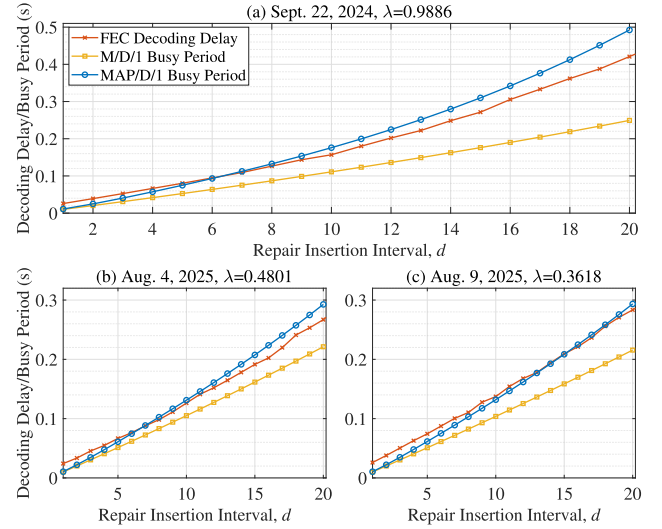


Fig. 12. Comparison of FEC decoding delay and MAP/D/1 busy period analysis; packet sending interval is 10 ms and $m = 16$ for MAP model.

We compare the number of states of the fitted statistical models in Table II. We see that MAP has many fewer states than LRL and RLRL. The number of states for LRL and RLRL is determined by the trace, whereas the order of MAP can be prespecified. A concise mathematical structure of MAP reduces the fitting complexity and facilitates performance analysis and FEC parameter design based on the model.

C. FEC Simulation and Busy Period

After verifying that MAP captures Starlink's packet loss behavior, this subsection analyzes the FEC decoding delay over Starlink connections. By modeling the delay as the busy period of a MAP/D/1 queue, it can be numerically computed using Algorithm 2. We use the fitted matrices $\{D_0, D_1\}$ of $m = 16$ from the previous subsection.

To validate the theoretical model, we conduct packet-level FEC transmissions on Ubuntu 20.04 (Intel i7-14700) via an emulated link using the original Starlink trace. The decoding delay is quantified as the time between the first source packet loss until its successful recovery, calculated by multiplying the total number of packets arriving at the buffer during this period with 10ms, which is the packet transmission interval corresponding to the trace measurement configuration (Section VI-A). Using traces collected in Canada on September 22, 2024, and in Germany on August 4 and 9, 2025, respectively, the average delay across multiple decoder activations are compared with the numerically computed busy period for various repair packet insertion intervals d in Fig. 12.

Fig. 12 shows that the MAP/D/1 busy periods demonstrate good consistency with the simulated FEC decoding delays with some minor deviations. We point out that the deviations are primarily due to disparities between model assumptions and actual system behaviors in the following two aspects. First, the model implicitly assumes that all repair packets can be successfully received by assuming a constant queue service time. Given repair packet insertion rate $f = (d-1)/d$, the effective insertion probability is actually $f(1-\varepsilon)$ where

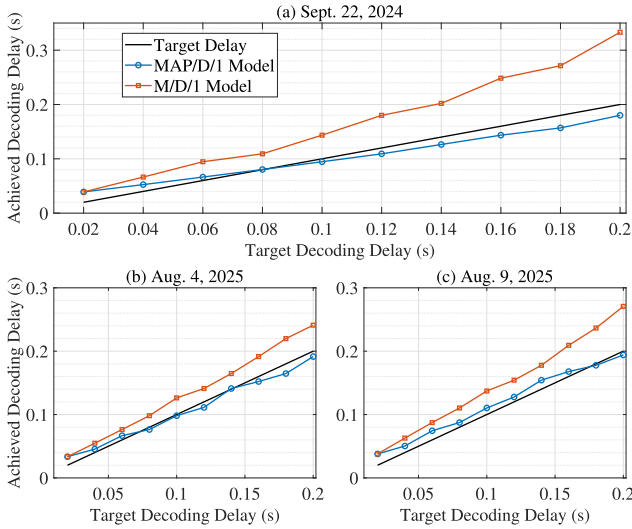


Fig. 13. Comparative evaluation of target versus achieved decoding delay; packet sending interval is 10 ms and $m = 16$ for MAP model.

ε denotes the packet loss rate. This would lead to delay underestimation. Second, the model assumes that a lost source packet must undergo a constant repair sending interval, i.e., the time proportional to sending d packets, such that the corresponding row in the decoding matrix can be filled in by a repair packet. However, in fact the corresponding row of the first lost source packet is filled in by the next repair packet, which usually arrives within less than d packets. This would cause delay overestimation. These two bias mechanisms coexist in the system, but their dominance varies with the repair packet insertion interval. A small interval primarily leads to delay underestimation, because the repair insertion rate is high (and hence ignoring repair packet loss causes more deviation), whereas delay overestimation dominates if the insertion interval is long.

In Fig. 12, we have also plotted busy periods of an M/D/1 queue, where its arrival rate is set to be equal to the average loss rate of the Starlink trace. This corresponds to the decoding delay under the assumption that the loss process is Bernoulli [19]. The result shows that, if we had mistakenly modeled the loss pattern of LEO scenario (say as Bernoulli), we may significantly underestimate the delay for the same repair packet sending interval. This intuitively demonstrates the motivation of the MAP modeling of loss behavior and the FEC delay analysis, which offer more meaningful guidelines for FEC design in LEO satellite networks.

To demonstrate the practical applicability of our analytical framework, we present a case study. This study employs the proposed MAP-based packet loss modeling and virtual queue analysis method to determine optimal FEC repair packet insertion intervals under specified decoding delay constraints. Our methodology employs a systematic search algorithm that evaluates candidate insertion intervals through estimated decoding delay computation, progressively refining the selection until convergence to the optimal value.

As shown in Fig. 13, the decoding delays resulting from the repair packet insertion intervals estimated by our proposed

model exhibit small deviations from target values across datasets collected in Victoria, Canada (September 22, 2024) and Brühl, Germany (August 4 and 9, 2025). A comparison with M/D/1-based analysis shows that our method's estimates achieve significantly closer agreement with actual system requirements. This case study validates the capability of our model in optimizing FEC performance in practical LEO satellite networks.

VII. DISCUSSIONS AND FUTURE WORK

LEO satellite networks have attracted wide attention these days. In this section, we discuss several limitations of the work and point out future research directions. First, while the proposed model has been validated using data from bent-pipe connections within the Starlink network, its applicability to LEO satellite systems incorporating inter-satellite links (ISL) or distinct orbital configurations requires further investigation. Future research will therefore focus on adapting the modeling framework to accommodate these architectures.

Second, while this work employs MAP to model packet loss over a fixed time period, the dynamic nature of LEO networks necessitates adaptive parameter updates to maintain model accuracy. Developing incremental or online learning algorithms to continuously refine the MAP parameters in real time remains an open challenge for practical deployment.

Third, we note that the inspiration of using MAP for modeling the loss of LEO satellite networks was because the losses may be incurred by periodic handover, intermittent blockage due to cloud or rain [44] and so on, which may be governed by some hidden state transitions of varying rates. These naturally fit the structure of MAP. However, currently we have only followed an end-to-end approach. That means we treat various factors of weather conditions, peak usage hours, dish orientation, location, or even geomagnetic storms [45] and solar flares [46] as latent variables, only inferring their influence indirectly through observed packet loss. Explicitly parameterizing these physical mechanisms is of great interest, and has good potential for further improving model interpretability and accuracy.

Finally, this paper considers separately modeling loss pattern and then analyzing/designing FEC parameters. This may be sub-optimal given that the underlying loss process of LEO satellites being highly dynamic. Adjusting FEC parameters on-the-fly given observations of the process leveraging reinforcement learning techniques, which jointly track loss dynamics and make FEC repair decisions, is also an interesting problem.

VIII. CONCLUSION

In this paper, we have shown that existing packet loss models are insufficient to characterize the packet loss patterns of LEO satellite networks. We have proposed to model the packet loss using MAP, and show that much better fitting performance can be achieved as compared to traditional models. Based on the MAP model, we have revealed that the busy period of MAP/D/1 queue can approximate the decoding delay of a streaming FEC scheme quite well.

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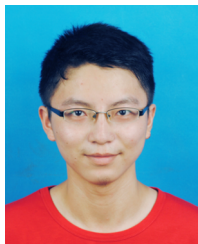
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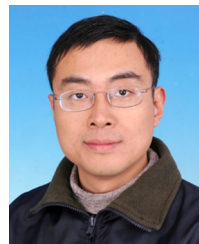
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